However, we now have

$$\int_0^a e^{-sx} x^m \, dx = \int_0^\infty e^{-sx} x^m \, dx - \int_a^\infty e^{-sx} x^m \, dx.$$

It is clear from (7), and from the fact that $m < 0$, that if we let $N \to \infty$, then $\int_a^\infty e^{-sx} x^{m-1} \, dx = O(1/|s|)$. The proposition if therefore proved.

Note. If one wants to obtain a better error term in Proposition 1.3, or for that matter extend the range of m, then one needs to mitigate the effect of the contribution of the end-point $x = a$. This can be done by introducing suitable smooth cut-offs. See Problem 1.

2 Laplace's method; Stirling's formula

We have already mentioned that when Φ is real-valued, the main contribution to $\int_a^b e^{-s\Phi(x)} \, dx$ as $s \to \infty$ comes from the point where Φ takes its minimum value. A situation where this minimum is attained at one of the end-points, a or b, was considered in Proposition 1.2. We now turn to the important case when the minimum is achieved in the interior of $[a, b]$.

Consider

$$\int_a^b e^{-s\Phi(x)} \psi(x) \, dx$$

where the **phase** Φ is real-valued, and both it and the **amplitude** ψ are assumed for simplicity to be indefinitely differentiable. Our hypothesis regarding the minimum of Φ is that there is an $x_0 \in (a, b)$ so that $\Phi'(x_0) = 0$, but $\Phi''(x_0) > 0$ throughout $[a, b]$ (Figure 2 illustrates the situation.)

Proposition 2.1 *Under the above assumptions, with $s > 0$ and $s \to \infty$,*

$$\int_a^b e^{-s\Phi(x)} \psi(x) \, dx = e^{-s\Phi(x_0)} \left[\frac{A}{s^{1/2}} + O\left(\frac{1}{s}\right) \right], \tag{8}$$

where

$$A = \sqrt{2\pi} \frac{\psi(x_0)}{(\Phi''(x_0))^{1/2}}.$$

Proof. By replacing $\Phi(x)$ by $\Phi(x) - \Phi(x_0)$ we may assume that $\Phi(x_0) = 0$. Since $\Phi'(x_0) = 0$, we note that

$$\frac{\Phi(x)}{(x - x_0)^2} = \frac{\Phi''(x_0)}{2} \varphi(x),$$

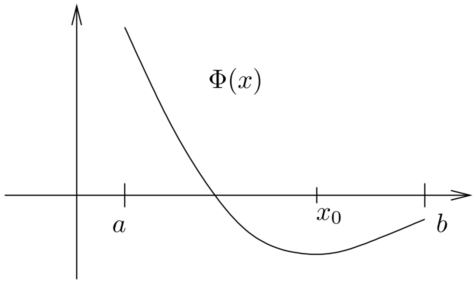

Figure 2. The function Φ, with its minimum at x_0

where φ is smooth, and $\varphi(x) = 1 + O(x - x_0)$ as $x \to x_0$. We can therefore make the smooth change of variables $x \mapsto y = (x - x_0)(\varphi(x))^{1/2}$ in a small neighborhood of $x = x_0$, and observe that $dy/dx|_{x_0} = 1$, and thus $dx/dy = 1 + O(y)$ as $y \to 0$. Moreover, we have $\psi(x) = \tilde{\psi}(y)$ with $\tilde{\psi}(y) = \psi(x_0) + O(y)$ as $y \to 0$. Hence if $[a', b']$ is a sufficiently small interval containing x_0 in its interior, by making the indicated change of variables we obtain

(9)

$$\int_{a'}^{b'} e^{-s\Phi(x)}\psi(x)\,dx = \psi(x_0)\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}\,dy + O\left(\int_\alpha^\beta e^{-s\frac{\Phi''(x_0)}{2}y^2}|y|dy\right),$$

where $\alpha < 0 < \beta$. We now make the further change of variables $y^2 = X$, $dy = \frac{1}{2}X^{-1/2}\,dX$, and we see by (5) that the first integral on the right-hand side in (9) is

$$\int_0^{a_0} e^{-s\frac{\Phi''(x_0)}{2}X} X^{-1/2}\,dX + O(e^{-\delta s}) = s^{-1/2}\left(\frac{2\pi}{\Phi''(x_0)}\right)^{1/2} + O(e^{-\delta s}),$$

for some $\delta > 0$. By the same argument, the second integral is $O(1/s)$. What remains are the integrals of $e^{-s\Phi(x)}\psi(x)$ over $[a, a']$ and $[b', b]$; but these integrals decay exponentially as $s \to \infty$, since $\Phi(x) \geq c > 0$ in these two sub-intervals. Altogether, this establishes (8) and the proposition.

It is important to realize that the asymptotic relation (8) extends to all complex s with $\mathrm{Re}(s) \geq 0$. The proof, however, requires a somewhat different argument: here we must take into account the oscillations of $e^{-s\Phi(x)}$ when $|s|$ is large but $\mathrm{Re}(s)$ is small, and this is achieved by a simple integration by parts.

Proposition 2.2 *With the same assumptions on* Φ *and* ψ*, the relation* (8) *continues to hold if* $|s| \to \infty$ *with* $\mathrm{Re}(s) \geq 0$.

Proof. We proceed as before to the equation (9), and obtain the appropriate asymptotic for the first term, by virtue of Proposition 1.3, with $m = -1/2$. To deal with the rest we start with an observation. If Ψ and ψ are given on an interval $[\overline{a}, \overline{b}]$, are indefinitely differentiable, and $\Psi(x) \geq 0$, while $|\Psi'(x)| \geq c > 0$, then if $\mathrm{Re}(s) \geq 0$,

$$\int_{\overline{a}}^{\overline{b}} e^{-s\Psi(x)}\psi(x)\,dx = O\left(\frac{1}{|s|}\right) \qquad \text{as } |s| \to \infty. \tag{10}$$

Indeed, the integral equals

$$-\frac{1}{s}\int_{\overline{a}}^{\overline{b}} \frac{d}{dx}\left(e^{-s\Psi(x)}\right)\frac{\psi(x)}{\Psi'(x)}\,dx,$$

which by integration by parts gives

$$\frac{1}{s}\int_{\overline{a}}^{\overline{b}} e^{-s\Psi(x)}\frac{d}{dx}\left(\frac{\psi(x)}{\Psi'(x)}\right)\,dx - \frac{1}{s}\left[e^{-s\Psi(x)}\frac{\psi(x)}{\Psi'(x)}\right]_{\overline{a}}^{\overline{b}}.$$

The assertion (10) follows immediately since $|e^{-s\Psi(x)}| \leq 1$, when $\mathrm{Re}(s) \geq 0$. This allows us to deal with the integrals of $e^{-s\Phi(x)}\psi(x)$ in the complementary intervals $[a, a']$ and $[b', b]$, because in each, $|\Phi'(x)| \geq c > 0$, since $\Phi'(x_0) = 0$ and $\Phi''(x) \geq c_1 > 0$.

Finally, for the second term on the right-hand side of (9) we observe that it is actually of the form

$$\int_{\alpha}^{\beta} e^{-s\frac{\Phi''(x_0)}{2}y^2}y\eta(y)\,dy,$$

where $\eta(y)$ is differentiable. Then we can again estimate this term by integration by parts, once we write it as

$$-\frac{1}{s\Phi''(x_0)}\int_{\alpha}^{\beta}\frac{d}{dy}\left(e^{-s\frac{\Phi''(x_0)}{2}y^2}\right)\eta(y)\,dy,$$

obtaining the bound $O(1/|s|)$.

The special case of Proposition 2.2 when s is purely imaginary, $s = it$, $t \to \pm\infty$, is often treated separately; the argument in this situation is usually referred to as the method of **stationary phase**. The points x_0 for which $\Phi'(x_0) = 0$ are called the **critical points**.

Our first application will be to the asymptotic behavior of the gamma function Γ, given by Stirling's formula. This formula will be valid in any sector of the complex plane that omits the negative real axis. For any $\delta > 0$ we set $S_\delta = \{s : |\arg s| \leq \pi - \delta\}$, and denote by $\log s$ the principal branch of the logarithm that is given in the plane slit along the negative real axis.

Theorem 2.3 *If $|s| \to \infty$ with $s \in S_\delta$, then*

$$\Gamma(s) = e^{s \log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \left(1 + O\left(\frac{1}{|s|^{1/2}}\right)\right). \tag{11}$$

Remark. With a little extra effort one can improve the error term to $O(1/|s|)$, and in fact obtain a complete asymptotic expansion in powers of $1/s$; see Problem 2. Also, we note that (11) implies $\Gamma(s) \sim \sqrt{2\pi} s^{s-1/2} e^{-s}$, which is how Stirling's formula is often stated.

To prove the theorem we first establish (11) in the right half-plane. We shall show that the formula holds whenever $\mathrm{Re}(s) > 0$, and in addition that the error term is uniform on the closed half-plane, once we omit a neighborhood of the origin (say $|s| < 1$). To see this, start with $s > 0$, and write

$$\Gamma(s) = \int_0^\infty e^{-x} x^s \, \frac{dx}{x} = \int_0^\infty e^{-x + s \log x} \, \frac{dx}{x}.$$

Upon making the change of variables $x \mapsto sx$, the above equals

$$\int_0^\infty e^{-sx + s \log sx} \, \frac{dx}{x} = e^{s \log s} e^{-s} \int_0^\infty e^{-s\Phi(x)} \, \frac{dx}{x},$$

where $\Phi(x) = x - 1 - \log x$. By analytic continuation this identity continues to hold, and we have when $\mathrm{Re}(s) > 0$,

$$\Gamma(s) = e^{s \log s} e^{-s} I(s)$$

with

$$I(s) = \int_0^\infty e^{-s\Phi(x)} \, \frac{dx}{x}.$$

It now suffices to see that

$$I(s) = \frac{\sqrt{2\pi}}{s^{1/2}} + O\left(\frac{1}{|s|}\right) \qquad \text{for} \qquad \mathrm{Re}(s) > 0. \tag{12}$$

Observe first that $\Phi(1) = \Phi'(1) = 0$, $\Phi''(x) = 1/x^2 > 0$ whenever $0 < x < \infty$, and $\Phi''(1) = 1$. Thus Φ is convex, attains its minimum at $x = 1$, and is positive.

We apply the complex version of the Laplace method, Proposition 2.2, in this situation. Here the critical point is $x_0 = 1$ and $\psi(x) = 1/x$. We choose for convenience the interval $[a, b]$ to be $[1/2, 2]$. Then for $\int_a^b e^{-s\Phi(x)}\psi(x)\,dx$ we get the asymptotic (12). It remains to bound the error terms, those corresponding to integration over $[0, 1/2]$, and $[2, \infty)$. Here the device of integration by parts, which has served us so well, can be applied again. Indeed, since $\Phi'(x) = 1 - 1/x$, we have

$$\begin{aligned}\int_\epsilon^{1/2} e^{-s\Phi(x)}\,\frac{dx}{x} &= -\frac{1}{s}\int_\epsilon^{1/2} \frac{d}{dx}\left(e^{-s\Phi(x)}\right)\frac{dx}{\Phi'(x)x}\\ &= -\frac{1}{s}\left[\frac{e^{-s\Phi(x)}}{x-1}\right]_\epsilon^{1/2} - \frac{1}{s}\int_\epsilon^{1/2} e^{-s\Phi(x)}\frac{dx}{(x-1)^2}.\end{aligned}$$

Noting that $\Phi(\epsilon) \to +\infty$ as $\epsilon \to 0$, and $|e^{-s\Phi(x)}| \le 1$, we find in the limit that

$$\int_0^{1/2} e^{-s\Phi(x)}\,\frac{dx}{x} = \frac{2}{s}e^{-s\Phi(1/2)} - \frac{1}{s}\int_0^{1/2} e^{-s\Phi(x)}\frac{dx}{(x-1)^2}.$$

Thus the left-hand side is $O(1/|s|)$ in the half-plane $\mathrm{Re}(s) \ge 0$.

The integral $\int_2^\infty e^{-s\Phi(x)}\,\frac{dx}{x}$ is treated analogously, once we note that $\int_2^\infty (x-1)^{-2}\,dx$ converges.

Since these estimates are uniform, (12) and thus (11) are proved for $\mathrm{Re}(s) \ge 0$, $|s| \to \infty$.

To pass from $\mathrm{Re}(s) \ge 0$ to $\mathrm{Re}(s) \le 0$, $s \in S_\delta$, we record the following fact about the principal branch of $\log s$: whenever $\mathrm{Re}(s) \ge 0$, $s = \sigma + it$, $t \ne 0$, then

$$\log(-s) = \begin{cases} \log s - i\pi & \text{if } t > 0,\\ \log s + i\pi & \text{if } t < 0.\end{cases}$$

Hence if $G(s) = e^{s\log s}e^{-s}$, $\mathrm{Re}(s) \ge 0$, $t \ne 0$, then

$$G(-s)^{-1} = \begin{cases} e^{s\log s}e^{-s}e^{-si\pi} & \text{if } t > 0,\\ e^{s\log s}e^{-s}e^{si\pi} & \text{if } t < 0.\end{cases} \tag{13}$$

Next,

$$\Gamma(s)\Gamma(-s) = \frac{\pi}{-s\sin \pi s}, \tag{14}$$

which follows from the fact that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$, and $\Gamma(1-s) = -s\Gamma(-s)$ (see Theorem 1.4 and Lemma 1.2 in Chapter 6). The combination of (13) and (14), together with the fact that for large s, $\left(1 + O(1/|s|^{1/2})\right)^{-1} = 1 + O(1/|s|^{1/2})$, allows us then to extend (11) to the whole sector S_δ, thereby completing the proof of the theorem.

3 The Airy function

The Airy function appeared first in optics, and more precisely, in the analysis of the intensity of light near a caustic; it was an important early instance in the study of asymptotics of integrals, and it continues to arise in a number of other problems. The **Airy function** Ai is defined by

$$\text{(15)} \qquad \mathrm{Ai}(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(x^3/3+sx)}\, dx, \qquad \text{with } s \in \mathbb{R}.$$

Let us first see that because of the rapid oscillations of the integrand as $|x| \to \infty$, the integral converges and represents a continuous function of s. In fact, note that

$$\frac{1}{i(x^2+s)} \frac{d}{dx} \left(e^{i(x^3/3+sx)} \right) = e^{i(x^3/3+sx)},$$

so if $a \geq 2|s|^{1/2}$, we can write the integral $\int_a^R e^{i(x^3/3+sx)}\, dx$ as

$$\text{(16)} \qquad \int_a^R \frac{1}{i(x^2+s)} \frac{d}{dx} \left(e^{i(x^3/3+sx)} \right) dx.$$

We may now integrate by parts and let $R \to \infty$, to see that the integral converges uniformly, and that as a result $\int_a^\infty e^{i(x^3/3+sx)}\, dx$ is also continuous for $|s| \leq a^2/4$. The same argument works for the integral from $-\infty$ to $-a$ and our assertion regarding $\mathrm{Ai}(s)$ is established.

A better insight into $\mathrm{Ai}(s)$ is given by deforming the contour of integration in (15). A choice of an optimal contour will appear below, but for now let us notice that as soon as we replace the x-axis of integration in (15) by the parallel line $L_\delta = \{x + i\delta,\ x \in \mathbb{R}\}$, $\delta > 0$, matters improve dramatically.

In fact, we may apply the Cauchy theorem to $f(z) = e^{i(z^3/3+sz)}$ over the rectangle γ_R shown in Figure 3.

One observes that $f(z) = O(e^{-\delta x^2})$ on L_δ, while $f(z) = O(e^{-yR^2})$ on the vertical sides of the rectangle. Thus since $\int_0^\delta e^{-yR^2}\, dy \to 0$ as